

The Things Network (TTN)

The Things Network provides a set of **open tools** and a **global, open network** to build your next IoT application.

The free version is a **full access network**: all the free registered **Gateways are accessible**, this leverages the Network capabilities and coverage.

<https://www.thethingsnetwork.org/>



TTN Coverage Map

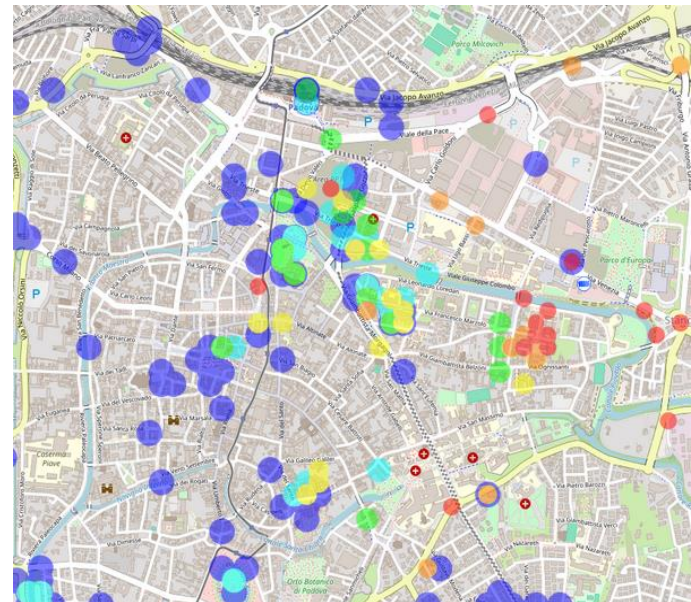
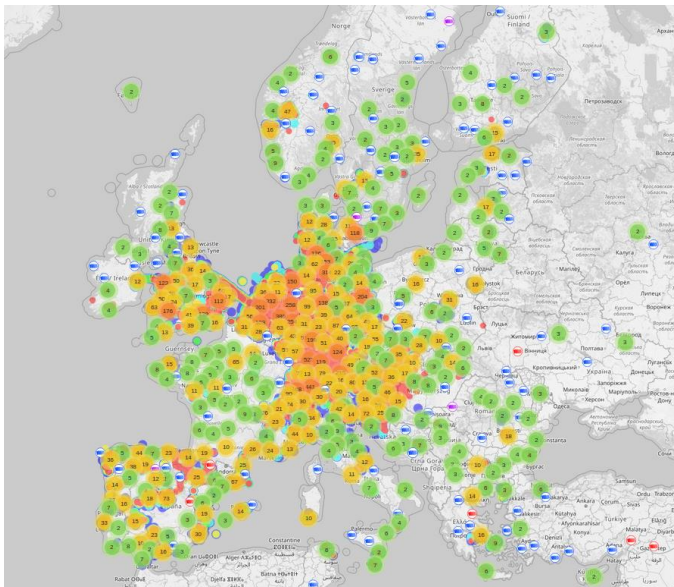
Gateways are expensive! 1 500-2000 €

Want to build your IoT project without a Gateway?

Check the TTN Coverage Map:

<https://ttnmapper.org/heatmap/>

FAIR USAGE POLICY!!!



- Once in TTN you can register for a free account
- Register Gateways (if any)
- Create an application
 - Register End-Devices:
 - JoinEUI (AppEUI): identifies Join Server, for now 0...0 (zeros)
 - DevEUI: provided by End-Device (Arduino Serial Monitor)
 - AppKey: provided by Network Server (TTN)
 - End-Device-ID: choose a string identifier for the device



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provaperiot

Application overview

End devices

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Top end devices



prova2

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prova2

ID: prova2

Device overview

Live data

TIME	TYPE	DATA PREVIEW
↑ 11:21:37	Join-request to cluster-local Join...	
↑ 11:21:30	Join-request to cluster-local Join...	
↑ 11:21:23	Join-request to cluster-local Join...	
↑ 11:21:02	Join-request to cluster-local Join...	
↑ 11:20:55	Join-request to cluster-local Join...	

1. Register device in TTN and check Live Data
 - (Both Arduino and TTN sides!)
2. Edit the Arduino code to send multiple messages after the join procedure has been completed.
(loop)



DIPARTIMENTO
DI INGEGNERIA
DELL'INFORMAZIONE